

Batch #43 - 25 Nov 2024

Refuse/Resist (Ukrainian Golden Ale) v3 - 8.3%

Ukrainian Golden Ale

Brewer: The Thirsty Otter

Type: All Grain

IBU : 25 (Tinseth)
 BU/GU : 0.34
 Colour : 15 EBC
 Carbonation : 2 CO2-vol

Pre-Boil Gravity : 1.036
 Original Gravity : 1.071
 Total Gravity : 1.073
 Final Gravity : 1.023

Fermentables (507 g)

469 g - Pale Ale Malt 10 EBC (92.5%)
 ^ Lot # (28.06.2022)
 ^ Brouwmaatje (NL) BM-BL.051.613.25/1
 23 g - Wheat Malt 5.5 EBC (4.5%)
 ^ Lot # (09.09.2021)
 ^ Brouwmaatje (NL) 051.125.3
 15 g - Chateau Biscuit 45 EBC (3%)
 ^ Lot # 5425000394839 (24.11.2022)
 7 g - Bottling - Sugar, Table (Sucrose) 2 EBC
 ^ Albert Heijn (NL)

Hops (4.6 g)

60 min - 1.3 g - Cascade (T90) - 7.5% (16 IBU)
 ^ Brouwmaatje (NL) BM-BL.053.148.3/100
 30 min - 0.9 g - Saaz - 3.6% (4 IBU)
 ^ Lot # T9020044SAA
 ^ Brouwmaatje (NL) BM-HUM.420000 Humlegardens
 www.humle.se
 15 min - 1.2 g - Saaz - 3.6% (3 IBU)
 ^ Lot # T9020044SAA
 ^ Brouwmaatje (NL) BM-HUM.420000 Humlegardens
 www.humle.se

Hop Stand

30 min hopstand @ 80 °C
 30 min - 1.2 g - Cascade (T90) - 7.5% (2 IBU)
 ^ Brouwmaatje (NL) BM-BL.053.148.3/100

02 PET Bottle 1.5L (60min) (rev 4)

Batch Size : 1.2 L
 Boil Size : 3.68 L
 Post-Boil Vol : 1.87 L

Mash Water : 1.52 L
 Sparge Water : 2.65 L
 Boil Time : 60 min
 Total Water : 4.17 L



15 EBC

Brewhouse Efficiency: 52.7%

Mash Efficiency: 79.1%

Mash Profile

04 High fermentability (60 min)
 77.8 °C - Strike Temp
 71 °C - 60 min - Temperature

Fermentation Profile

01 Ale + DR + Conditioning
 17 °C - 10 days - Primary
 21 °C - 4 days - Diacetyl rest
 18 °C - 14 days - Carbonation
 18 °C - 28 days - Conditioning

Water Profile

NL Spa Reine Flat Mineral Water (www.ah.nl) (...
 Ca 108 Mg 9 Na 8 Cl 128 SO 121 HCO 17

SO/Cl ratio: 1

Mash pH: 4.79

Sparge pH: 6

Measurements

Mash pH: 5.34

Boil Volume:

Pre-Boil Gravity: 1.050

Post-Boil Kettle Volume:

Original Gravity: 1.071

Fermenter Top-Up:

Fermenter Volume: 1.2

Final Gravity: 1.011

Bottling Volume: 0.9

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Miscellaneous

Mash - 0.489 g - Calcium Chloride (CaCl₂) 33 %
33%
^ Lot # 115038
^ Brouwstore (NL) 055.035.0
Mash - 0.011 g - Canning Salt (NaCl)
^ Albert Heijn (NL)
Mash - 0.062 g - Epsom Salt (MgSO₄)
^ Lot # /2119000091
^ Brouwstore (NL) 055.027.7
Mash - 0.144 g - Gypsum (CaSO₄)
^ Brouwmaatje (NL)
Mash - 0.129 ml - Lactic Acid 80% 80%
^ Lot # 20200213
^ Brouwstore (NL) 003.002.3
Mash - 1.111 l - NL Spa Reine Flat Mineral Water
^ AH (NL)
Sparge - 0.984 l - NL Spa Reine Flat Mineral
Water
^ AH (NL)
10 min - Boil - 0.214 g - Irish Moss
^ Lot # 4348
^ Brouwstore (NL)
5 min - Boil - 1.2 g - Coriander Seed
^ Brouwmaatje (NL)
Bottling - 3 items - BrewFerm Carbonation Drops
(33 cl)
^ Brouwmaatje (NL) BM-BL.007.009.33
Bottling - 3 items - 30 cl bottle (26 mm cap)
BNR-CBK
^ Bruin Nederlands Retour - Centraal Brouwerij
Kantoor, Netherlands
Bottling - 3 items - Bottle Caps 26 mm (Orange)
^ Lot # 694-201130-111430-176081-1/1
^ The Malt Miller (UK) EQU-14-007

Yeast

0.22 pkg - Fermentis SafAle English Ale S-04
^ Lot # 220210 2022 062
^ Brouwmaatje (NL) BM-BL.050.005.8

Recipe Notes

Add aroma hops and cool over night.

Batch Notes

Brew day
25-11-2024

Day 0 in fermentation

26-11-2024 09:14 pH = 5.07 @ 13.9 C.
26-11-2024 09:25 pitched 2.40 gram S-04 yeast.
26-11-2024 12:00 17.0 C.
26-11-2024 16:00 16.1 C.
26-11-2024 20:00 17.1 C.

Day 1 in fermentation

27-11-2024 00:00 17.5 C.
27-11-2024 04:00 17.5 C.
27-11-2024 08:00 17.5 C.

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27-11-2024 12:00 17.1 C., shake.
27-11-2024 20:00 17.0 C.

Day 2 in fermentation

28-11-2024 00:00 17.5 C.
28-11-2024 04:00 17.5 C.
28-11-2024 08:00 16.5 C.
28-11-2024 12:00 16.7 C., shake.
28-11-2024 16:00 17.1 C.
28-11-2024 20:00 17.3 C.

Day 3 in fermentation

29-11-2024 00:00 17.5 C.
29-11-2024 04:00 17.6 C.
29-11-2024 08:00 17.5 C.
29-11-2024 12:00 17.5 C.
29-11-2024 16:00 17.5 C.
29-11-2024 20:00 17.5 C.

Day 4 in fermentation

30-11-2024 00:00 17.5 C.
30-11-2024 04:00 17.4 C.
30-11-2024 08:00 17.1 C.
30-11-2024 12:00 16.8 C.
30-11-2024 16:00 17.4 C.
30-11-2024 20:00 17.0 C.

Day 5 in fermentation

01-12-2024 00:00 17.5 C.
01-12-2024 04:00 17.4 C.
01-12-2024 08:00 17.0 C.
01-12-2024 12:00 16.8 C.
01-12-2024 16:00 16.5 C.
01-12-2024 20:00 17.0 C.

Day 6 in fermentation

02-12-2024 00:00 17.3 C.
02-12-2024 04:00 17.5 C.
02-12-2024 08:00 17.5 C.
02-12-2024 12:00 17.4 C.
02-12-2024 16:00 17.5 C.
02-12-2024 20:00 17.5 C.

Day 7 in fermentation

03-12-2024 00:00 17.5 C.
03-12-2024 04:00 17.5 C.
03-12-2024 08:00 17.7 C.
03-12-2024 12:00 17.5 C.
03-12-2024 16:00 17.5 C.
03-12-2024 20:00 17.5 C.

Day 8 in fermentation

04-12-2024 00:00 17.5 C.
04-12-2024 03:45 17.5 C.
04-12-2024 08:00 17.2 C.
04-12-2024 12:00 17.1 C.
04-12-2024 16:00 17.0 C.
04-12-2024 20:00 16.4 C.
04-12-2024 23:45 16.7 C.

Day 9 in fermentation

05-12-2024 04:00 16.5 C.
05-12-2024 08:00 16.5 C.
05-12-2024 12:00 16.7 C.
05-12-2024 16:00 16.9 C.
05-12-2024 20:00 16.9 C.

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Day 10 in fermentation

06-12-2024 00:00 17.1 C.
06-12-2024 04:00 16.9 C.
06-12-2024 08:15 17.5 C.
06-12-2024 12:00 17.1 C.
06-12-2024 16:00 17.5 C.
06-12-2024 20:00 19.8 C.

Day 11 in fermentation

07-12-2024 01:00 21.0 C.
07-12-2024 04:00 21.0 C.
07-12-2024 08:00 21.0 C.
07-12-2024 12:00 20.8 C.
07-12-2024 16:00 20.5 C.
07-12-2024 20:00 20.1 C.

Day 12 in fermentation

08-12-2024 00:00 20.0 C.
08-12-2024 04:00 20.1 C.
08-12-2024 08:00 20.4 C.
08-12-2024 12:00 20.5 C.
08-12-2024 16:00 20.7 C.
08-12-2024 20:00 21.0 C.

Day 13 in fermentation

09-12-2024 00:00 20.7 C.
09-12-2024 04:00 20.5 C.
09-12-2024 08:00 20.2 C.
09-12-2024 12:00 20.2 C.
09-12-2024 16:00 20.0 C.
09-12-2024 20:00 20.0 C.

Day 14 in fermentation

10-12-2024 00:00 20.0 C.
10-12-2024 04:00 20.1 C.
10-12-2024 08:00 20.4 C.
10-12-2024 12:00 20.9 C.
10-12-2024 16:00 21.1 C.
10-12-2024 20:00 21.1 C.

Day 15 in fermentation

11-12-2024 00:00 21.0 C.
11-12-2024 04:00 20.8 C.
11-12-2024 08:00 20.4 C.
11-12-2024 12:00 20.2 C.
11-12-2024 16:00 20.0 C.
11-12-2024 20:00 20.0 C.

Day 16 in fermentation

12-12-2024 00:00 20.0 C.
12-12-2024 04:00 20.1 C.
12-12-2024 08:00 20.3 C.
12-12-2024 12:00 20.7 C.
12-12-2024 16:00 21.0 C.
12-12-2024 20:00 21.0 C.

Day 17 in fermentation

13-12-2024 00:00 20.9 C.
13-12-2024 04:00 20.9 C.
13-12-2024 08:00 20.3 C.
13-12-2024 12:00 20.0 C.
13-12-2024 16:00 20.0 C.

Bottling day

13-12-2024 19:21 15.9 C., SG = 1.011, pH = 4.24.

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Batch Notes

Day 0 in conditioning

13-12-2024 20:00 19.9 C.

Day 1 in conditioning

14-12-2024 00:00 17.3 C.

14-12-2024 04:00 17.5 C.

14-12-2024 08:00 17.4 C.

14-12-2024 12:00 16.9 C.

14-12-2024 16:00 16.8 C.

14-12-2024 20:00 16.5 C.

Day 2 in conditioning

15-12-2024 02:30 17.1 C.

15-12-2024 04:00 16.5 C.

15-12-2024 08:00 16.9 C.

15-12-2024 12:00 17.3 C.

15-12-2024 16:00 13.2 C.

15-12-2024 20:00 16.8 C.

Day 3 in conditioning

16-12-2024 00:00 16.9 C.

16-12-2024 04:00 17.3 C.

16-12-2024 08:00 17.5 C.

16-12-2024 12:00 17.1 C.

16-12-2024 16:00 16.7 C.

16-12-2024 20:00 16.8 C.

Day 4 in conditioning

17-12-2024 00:00 17.5 C.

17-12-2024 04:00 17.0 C.

17-12-2024 08:00 16.5 C.

17-12-2024 12:00 17.2 C.

17-12-2024 16:00 17.1 C.

17-12-2024 19:45 16.7 C.

Day 5 in conditioning

18-12-2024 00:00 17.4 C.

18-12-2024 04:00 17.1 C.

18-12-2024 08:00 16.5 C.

18-12-2024 12:00 17.5 C.

18-12-2024 16:00 16.7 C.

18-12-2024 20:00 16.7 C.

Day 6 in conditioning

19-12-2024 00:00 17.5 C.

19-12-2024 04:00 17.2 C.

19-12-2024 08:00 16.7 C.

19-12-2024 12:00 16.8 C.

19-12-2024 16:00 17.5 C.

19-12-2024 20:00 17.0 C.

Day 7 in conditioning

20-12-2024 00:00 16.5 C.

20-12-2024 04:00 17.1 C.

20-12-2024 08:00 17.4 C.

20-12-2024 12:00 17.2 C.

20-12-2024 16:00 16.5 C.

20-12-2024 20:00 16.8 C.

Day 8 in conditioning

21-12-2024 00:00 17.4 C.

21-12-2024 04:00 16.7 C.

21-12-2024 08:00 17.1 C.

21-12-2024 12:00 17.4 C.

21-12-2024 16:00 16.4 C.

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21-12-2024 20:00 17.0 C.

Day 9 in conditioning

22-12-2024 00:00 17.6 C.
22-12-2024 04:00 16.6 C.
22-12-2024 08:00 16.9 C.
22-12-2024 12:00 17.5 C.
22-12-2024 16:00 17.0 C.
22-12-2024 20:00 16.5 C.

Day 10 in conditioning

23-12-2024 00:00 16.6 C.
23-12-2024 04:00 16.8 C.
23-12-2024 08:00 17.0 C.
23-12-2024 12:00 17.2 C.
23-12-2024 16:00 17.4 C.
23-12-2024 20:00 17.4 C.

Day 11 in conditioning

24-12-2024 00:00 17.5 C.
24-12-2024 04:00 17.6 C.
24-12-2024 08:00 17.5 C.
24-12-2024 12:00 17.6 C.
24-12-2024 15:45 17.5 C.
24-12-2024 20:00 17.1 C.

Day 12 in conditioning

25-12-2024 00:00 16.5 C.
25-12-2024 04:00 16.8 C.
25-12-2024 08:00 17.2 C.
25-12-2024 12:00 17.5 C.
25-12-2024 16:00 17.2 C.
25-12-2024 20:00 16.7 C.

Day 13 in conditioning

26-12-2024 00:00 16.5 C.
26-12-2024 04:00 17.0 C.
26-12-2024 08:00 17.4 C.
26-12-2024 12:00 17.5 C.
26-12-2024 16:00 16.8 C.
26-12-2024 20:00 16.6 C.

Day 14 in conditioning

27-12-2024 00:00 17.1 C.
27-12-2024 04:15 17.5 C.
27-12-2024 08:00 17.4 C.
27-12-2024 12:00 16.6 C.
27-12-2024 16:00 16.5 C.
27-12-2024 20:00 16.7 C.

Day 15 in conditioning

28-12-2024 00:00 16.8 C.
28-12-2024 04:00 16.9 C.
28-12-2024 08:00 16.9 C.
28-12-2024 12:00 16.8 C.
28-12-2024 16:00 16.8 C.
28-12-2024 20:00 16.8 C.

Day 16 in conditioning

29-12-2024 00:00 16.6 C.
29-12-2024 04:00 16.5 C.
29-12-2024 08:00 16.8 C.
29-12-2024 12:00 17.1 C.
29-12-2024 16:00 16.8 C.
29-12-2024 20:00 16.7 C.

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Day 17 in conditioning

30-12-2024 00:00 16.5 C.
30-12-2024 04:00 16.5 C.
30-12-2024 08:00 16.8 C.
30-12-2024 12:00 17.2 C.
30-12-2024 16:00 17.3 C.
30-12-2024 20:00 17.4 C.

Day 18 in conditioning

31-12-2024 00:00 17.4 C.
31-12-2024 04:00 16.7 C.
31-12-2024 08:00 16.7 C.
31-12-2024 12:00 17.0 C.
31-12-2024 16:00 17.0 C.
31-12-2024 20:00 17.1 C.

Day 19 in conditioning

01-01-2025 00:00 17.2 C.
01-01-2025 04:00 17.4 C.
01-01-2025 08:00 17.0 C.
01-01-2025 12:00 16.5 C.
01-01-2025 16:00 16.7 C.
01-01-2025 20:00 17.0 C.

Day 20 in conditioning

02-01-2025 00:00 17.1 C.
02-01-2025 04:00 17.3 C.
02-01-2025 08:00 17.4 C.
02-01-2025 12:00 17.3 C.
02-01-2025 16:00 17.2 C.
02-01-2025 20:00 17.3 C.

Day 21 in conditioning

03-01-2025 00:00 17.0 C.
03-01-2025 04:00 16.8 C.
03-01-2025 08:00 16.6 C.
03-01-2025 12:00 17.3 C.
03-01-2025 16:00 16.7 C.
03-01-2025 20:00 16.5 C.

Day 22 in conditioning

04-01-2025 00:00 17.5 C.
04-01-2025 04:00 17.2 C.
04-01-2025 08:00 16.9 C.
04-01-2025 12:00 16.8 C.
04-01-2025 16:00 17.4 C.
04-01-2025 20:00 17.2 C.

Day 23 in conditioning

05-01-2025 00:00 16.9 C.
05-01-2025 04:00 17.1 C.
05-01-2025 08:00 17.2 C.
05-01-2025 12:00 16.7 C.
05-01-2025 16:00 17.2 C.
05-01-2025 20:00 17.3 C.

Day 24 in conditioning

06-01-2025 00:00 17.5 C.
06-01-2025 04:00 16.9 C.
06-01-2025 08:00 16.8 C.
06-01-2025 12:00 17.5 C.
06-01-2025 16:00 16.6 C.
06-01-2025 20:00 16.9 C.

Day 25 in conditioning

07-01-2025 00:00 17.3 C.

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07-01-2025 04:00 17.5 C.
07-01-2025 08:00 17.5 C.
07-01-2025 12:00 17.3 C.
07-01-2025 16:00 16.8 C.
07-01-2025 20:00 16.8 C.

Day 26 in conditioning

08-01-2025 00:00 16.5 C.
08-01-2025 04:00 16.9 C.
08-01-2025 08:00 17.5 C.
08-01-2025 12:00 17.4 C.
08-01-2025 16:00 17.3 C.
08-01-2025 20:00 17.1 C.

Day 27 in conditioning

09-01-2025 00:00 16.9 C.
09-01-2025 04:00 16.9 C.
09-01-2025 08:00 16.5 C.
09-01-2025 12:00 17.3 C.
09-01-2025 16:00 17.1 C.
09-01-2025 20:00 16.7 C.

Day 28 in conditioning

10-01-2025 00:00 17.4 C.
10-01-2025 04:00 17.1 C.
10-01-2025 08:00 16.8 C.
10-01-2025 12:00 17.4 C.
10-01-2025 16:00 17.3 C.
10-01-2025 20:00 17.1 C.

Day 29 in conditioning

11-01-2025 00:00 16.5 C.
11-01-2025 04:00 17.2 C.
11-01-2025 08:00 17.4 C.
11-01-2025 12:00 16.5 C.
11-01-2025 16:00 17.1 C.
11-01-2025 20:00 16.7 C.

Day 30 in conditioning

12-01-2025 00:00 17.2 C.
12-01-2025 04:00 16.5 C.
12-01-2025 08:00 17.1 C.
12-01-2025 12:00 17.4 C.
12-01-2025 16:00 17.0 C.
12-01-2025 20:00 16.5 C.

Day 31 in conditioning

13-01-2025 00:00 17.3 C.
13-01-2025 04:00 16.9 C.
13-01-2025 08:00 17.3 C.
13-01-2025 12:00 16.7 C.
13-01-2025 16:00 17.3 C.
13-01-2025 20:00 16.7 C.

Day 32 in conditioning

14-01-2025 00:00 17.2 C.
14-01-2025 04:00 17.3 C.
14-01-2025 08:00 17.4 C.
14-01-2025 12:00 16.5 C.
14-01-2025 16:00 17.2 C.
14-01-2025 20:00 16.8 C.

Day 33 in conditioning

15-01-2025 00:00 16.5 C.
15-01-2025 04:00 16.6 C.
15-01-2025 08:00 16.9 C.

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15-01-2025 12:00 16.5 C.
15-01-2025 16:00 16.9 C.
15-01-2025 20:00 17.0 C.

Day 34 in conditioning

16-01-2025 00:00 17.3 C.
16-01-2025 04:00 16.7 C.
16-01-2025 08:00 16.5 C.
16-01-2025 12:00 17.0 C.
16-01-2025 16:00 17.1 C.
16-01-2025 20:00 17.0 C.

Day 35 in conditioning

17-01-2025 00:00 17.1 C.
17-01-2025 04:00 17.1 C.
17-01-2025 08:00 16.9 C.
17-01-2025 12:00 16.7 C.
17-01-2025 16:00 16.4 C.
17-01-2025 20:00 17.4 C.

Day 36 in conditioning

18-01-2025 00:00 17.1 C.
18-01-2025 04:00 16.7 C.
18-01-2025 08:00 17.2 C.
18-01-2025 12:00 16.5 C.
18-01-2025 16:00 17.1 C.
18-01-2025 20:00 17.4 C.

Day 37 in conditioning

19-01-2025 00:00 17.4 C.
19-01-2025 04:00 16.9 C.
19-01-2025 08:00 17.3 C.
19-01-2025 12:00 17.4 C.
19-01-2025 16:00 16.5 C.
19-01-2025 20:00 16.9 C.

Day 38 in conditioning

20-01-2025 00:00 17.2 C.
20-01-2025 04:00 17.3 C.
20-01-2025 08:00 17.3 C.
20-01-2025 16:00 17.0 C.
20-01-2025 20:00 17.2 C.

Day 39 in conditioning

21-01-2025 00:00 17.4 C.
21-01-2025 04:00 16.9 C.
21-01-2025 08:00 17.2 C.
21-01-2025 12:00 17.4 C.
21-01-2025 16:00 16.7 C.
21-01-2025 20:00 17.2 C.

Day 40 in conditioning

22-01-2025 00:00 17.3 C.
22-01-2025 04:00 17.4 C.
22-01-2025 08:00 17.3 C.
22-01-2025 12:00 16.9 C.
22-01-2025 16:00 17.2 C.
22-01-2025 20:00 16.8 C.

Day 41 in conditioning

23-01-2025 00:00 17.0 C.
23-01-2025 04:00 17.2 C.
23-01-2025 08:00 17.0 C.
23-01-2025 12:00 16.6 C.
23-01-2025 16:00 16.7 C.
23-01-2025 20:00 16.7 C.

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Day 42 in conditioning
24-01-2025 00:00 16.5 C.
24-01-2025 04:00 16.9 C.
24-01-2025 08:00 16.9 C.
24-01-2025 12:00 17.2 C.
24-01-2025 16:00 17.4 C.
24-01-2025 20:00 17.3 C.
24-01-2025 23:45 16.9 C.

Extra Measured Values

Strike Water Temperature	71
Sparge Water Temperature	71
Sparge Water pH	6.64
Strike Water pH	4.41
Pitch Temperature	13.9

Batch Log

25 November 2024	Brew Date
25 November 2024 11:00	Status: Brewing
25 November 2024 16:20	Split off 1.2 L wort from Batch #41.
26 November 2024 09:00	SV = 17.0 C.
26 November 2024 09:25	Status: Fermenting
26 November 2024	Fermentation Start
26 November 2024 16:38	SV = 17.5 C.
6 December 2024 17:35	SV = 21.0 C.
13 December 2024 19:21	Filled 3 BNR bottles (30 cl).
13 December 2024	Bottling Date
13 December 2024 20:50	SV = 17.5 C.
24 January 2025 23:59	Status: Completed

Taste

3.0 / 5.0